# The Missing Middle: A Statistical Investigation of the Exoplanet Radius Gap

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**Abstract**

Exoplanet demographics reveal a curious absence: relatively few planets are found with radii between roughly 1.5 and 2.0 Earth radii, a feature known as the radius gap, first reported by Fulton et al. (2017). This project examines the radius gap through a carefully constructed sample of 2,587 planets drawn from the NASA Exoplanet Archive, filtered to minimize the influence of measurement uncertainty and model-derived (rather than directly measured) parameters. Gaussian mixture modeling shows that a single Gaussian cannot describe the radius distribution, and kernel density estimation combined with bootstrap resampling locates a density minimum near $1.76\ R_{\oplus}$, stable in position across several independently filtered subsamples and across smoothing bandwidths. The feature is shallow, however: the density at the minimum lies only about $2\%$ below the adjacent lower-radius peak, far less pronounced than the deficit reported for the California-Kepler Survey sample. The two formal tests of multimodality disagree: Silverman's critical bandwidth test rejects unimodality ($p = 0.028$), while Hartigan's dip test does not ($p = 0.52$). A power simulation at this sample size accounts for the difference — the dip test recovers a Fulton-depth gap in every simulated realization but essentially never recovers a valley as shallow as this one, since its statistic responds to the depth of a valley rather than to the presence of a second mode. Turning to the planets themselves, we ask which properties distinguish those above the gap from those below it, deliberately excluding radius from consideration to avoid circular reasoning. Logistic regression and random forest classifiers, trained on orbital and stellar parameters alone, achieve moderate but meaningful predictive power (ROC-AUC of 0.77–0.79), with orbital period and insolation emerging as the most informative features. This period-dependence is quantified through classical hypothesis testing and through a Bayesian Beta-Binomial model, both of which find strong, threshold-robust evidence that planets on longer orbits are more likely to lie above the gap than their close-in counterparts. These results are consistent with atmospheric mass-loss theories, particularly photoevaporation and core-powered mass-loss, as physical drivers of the radius gap. Period-dependent transit detection bias, however, predicts the same signature, and the composite archive sample used here supports no completeness correction capable of separating the two.

## 1 Introduction

The past three decades of exoplanet discovery have transformed how we think about planetary systems. Where the solar system offers only eight examples to generalize from, the more than five thousand confirmed exoplanets now cataloged reveal statistical patterns that no single system could show on its own. One of the most striking of these patterns is a shortage of planets with radii between approximately 1.5 and 2.0 Earth radii — a feature that has come to be known as the radius gap, or sometimes the "radius valley."

The gap was first clearly resolved by Fulton et al. (2017) using precise radius measurements from the California-Kepler Survey, which showed that small, close-in planets cluster into two distinct populations: rocky super-Earths below the gap and gas-enveloped sub-Neptunes above it, with relatively few planets in

between. Two physical mechanisms have been proposed to explain this separation. The first, photoevaporation, holds that high-energy radiation from the host star strips away a planet's primordial hydrogen-helium envelope over the first hundred million years of its life, an effect that is strongest for planets orbiting close to their star. The second, core-powered mass-loss, proposes a similar outcome driven instead by the planet's own internal heat, acting more gradually over billions of years. While the two theories differ in timescale and underlying physics, both predict that a planet's orbital period should be closely tied to its final size — planets on shorter orbits are expected to lose more of their atmosphere and end up smaller, below the gap, while planets farther from their star retain thicker envelopes and remain above it. This shared prediction makes orbital period a natural point of departure for testing the gap observationally.

This project investigates the radius gap using data from the NASA Exoplanet Archive, structured around four questions. First, does the radius distribution of a carefully filtered planet sample show statistically robust evidence of bimodality, and where is the gap located? Second, do formal statistical tests of multimodality support this structure, or do they reveal the practical limits of such tests when applied to subtle features? Third, setting aside radius itself to avoid circular reasoning, can a planet's orbital and stellar properties predict which side of the gap it falls on, and which properties matter most? Fourth, is the period-dependence predicted by mass-loss theory borne out formally, both through classical hypothesis testing and through Bayesian estimation?

The remainder of this report proceeds as follows. Section 2 describes the data source and the filtering protocol used to construct the analysis sample. Section 3 presents the distributional modeling, gap location, and multimodality testing. Section 4 develops a classification approach to identify predictors of gap membership. Section 5 reports classical hypothesis tests comparing planet populations across period groups. Section 6 presents a Bayesian Beta-Binomial analysis of the same question. Section 7 discusses the findings in light of existing literature and outlines the limitations of this study, and Section 8 concludes.

# 2 Data and Methods

## 2.1 Data Source

The analysis is based on the Planetary Systems Composite Parameters (PSCompPars) table from the NASA Exoplanet Archive, downloaded on 2026-07-15. This table provides one row per confirmed planet, with each parameter drawn from the most complete or most recent literature source available — a "composite" design that maximizes coverage but means that different columns for the same planet may originate from different studies and are not guaranteed to be mutually self-consistent. The raw download contained 6,319 planets described by 84 columns, including orbital, physical, and stellar parameters together with their reported uncertainties.

## 2.2 Initial Exploration

Before filtering, an exploratory pass established several features of the raw data that shaped the filtering protocol described below. The radius distribution is strongly right-skewed (mean $5.84\ R_\oplus$ versus median $2.86\ R_\oplus$), motivating the use of $\log_{10}$(radius) throughout the analysis. A large valley near 7–9 $R_\oplus$ is visible in the raw histogram, but this reflects a pile-up of hot Jupiters rather than the much finer Fulton radius gap, which is expected between roughly $1.5$ and $2.0\ R_\oplus$. Orbital period is similarly skewed, with a sharp concentration near short periods (median $10.8$ days) reflecting detection bias: short-period planets are easier to detect via both transit (more frequent transits within a limited observing baseline) and radial velocity (larger reflex signal).

Discovery method breaks down as Transit (74%), Radial Velocity (19%), and nine remaining methods combined (7%). Orbital period differs sharply by method (Transit median $8.0$ days versus Radial Velocity

median 307.9 days; Mann-Whitney $p \approx 4.4 \times 10^{-300}$), and missingness in key parameters is strongly tied to discovery method rather than occurring at random: Microlensing and Pulsar Timing detections are missing 88–100% of stellar and orbital parameters, since these techniques cannot directly measure them, while Transit and Radial Velocity detections are nearly complete. This confirms that any filtering based on completeness will necessarily concentrate the sample toward Transit and Radial Velocity detections, and that this concentration reflects a genuine data-quality limitation rather than an arbitrary analytical choice.

A critical distinction exists between *measured* and *calculated* radius values. The archive derives some radii from the Chen & Kipping (2017) mass-radius relation rather than from direct transit or interferometric measurement, identifiable via the `pl_rade_reflink` field. Of the 6,319 planets, 1,518 (24%) have calculated rather than measured radii. Plotting mass against radius and coloring by this distinction shows that calculated values fall exactly on a smooth, deterministic piecewise curve, while measured values scatter naturally around it, reflecting genuine compositional diversity. Since a deterministic formula cannot, by construction, reproduce fine distributional structure such as the radius gap, including calculated radii would artificially smooth out the very feature under investigation. Cross-tabulation with mass provenance further showed that 94.6% of radial-velocity-derived (Msini) mass entries carry a calculated radius, confirming that this is not a marginal concern.

A final exploratory result concerns system architecture rather than data quality, and is referred to later in Sections 6.4 and 7.2. Among the 1,037 multi-planet systems with at least two measured radii, the mean within-system standard deviation of $\log_{10}$(radius) is 0.135. Reassigning radii at random across systems while holding the system-size structure fixed (300 permutations) gives a null mean of 0.220, with no permutation reaching a value as low as the observed one ($p < 0.0033$). Planets within a system are therefore markedly more similar in size than randomly assembled groups of the same sizes, an independent confirmation of the "peas in a pod" effect of Weiss et al. (2018). This also means planetary radii are not independent between planets sharing a host, a point taken up in Section 6.4.

## 2.3 Filtering Protocol

Based on these findings, planets were retained through a sequential cascade of quality filters, summarized in Table 1. Each step addresses a specific, pre-specified concern: missing core variables, disputed confirmation status, calculated (non-measured) radius or period values, limit values that are not true point measurements, and finally the target radius range with a bound on relative measurement error.

Table 1: Filtering cascade from the raw sample to the main analysis sample.

| Step | Filter | N remaining | N removed |
|---|---|---|---|
| 0 | Start (raw PSCompPars) | 6319 | – |
| 1 | Has `pl_rade` and `pl_orbper` | 5929 | 390 |
| 2 | Not controversial (`pl_controv_flag`=0) | 5879 | 50 |
| 3 | Radius measured, not calculated | 4713 | 1166 |
| 4 | Radius not a limit value | 4710 | 3 |
| 5 | Period not a limit value | 4710 | 0 |
| 6 | Radius in 0.5–4 $R_\oplus$ | 3512 | 1198 |
| 7 | Relative radius error $< 20\%$ | **2587** | 925 |

The radius range applied at Step 6 (0.5–4 $R_\oplus$) follows Fulton et al. (2017) and focuses the sample on the region where the radius gap is expected. The relative-error threshold at Step 7 ($< 20\%$) was chosen because measurement noise of this magnitude is comparable to the width of the gap itself (roughly 0.3–0.5 $R_\oplus$) and

could obscure it entirely if left unfiltered. These two steps remove the largest numbers of planets (1198 and 925, respectively) and are therefore the primary candidates for sensitivity testing, discussed in Section 3.2.

This cascade yields a main analysis sample of **2,587 planets**. As a documented sensitivity variant, restricting Step 7 to Transit-detected planets only yields 2,562 planets — a difference of just 25 planets, since the preceding quality filters had already implicitly removed nearly all non-Transit detections (most of which carried calculated, not measured, radii). This indicates that explicit restriction to a single discovery method has little additional effect once quality filtering is applied, and this variant is retained specifically for the sensitivity analysis in Section 3.2 rather than as part of the primary filtering cascade. An orbital period restriction (e.g., $< 100$ days, following Fulton et al. 2017) was deliberately not applied at this stage in order to preserve statistical power; it is instead tested as one of several documented sensitivity variants in Section 3.2.

# 3 Distributional Modeling, Gap Location, and Multimodality Testing

## 3.1 Gaussian Mixture Modeling

To formally test whether the radius distribution is bimodal, Gaussian Mixture Models (GMMs) with one, two, and three components were fitted to $\log_{10}$(radius) and compared using the Akaike and Bayesian Information Criteria (AIC, BIC), which balance model fit against complexity. Table 2 summarizes the results.

Table 2: GMM model comparison across component counts.

| Components | AIC | BIC |
|---|---|---|
| 1 | $-1724.37$ | $-1712.65$ |
| 2 | $-1965.93$ | $-1936.63$ |
| 3 | $-1985.55$ | $-1938.68$ |

Both criteria improve substantially from one to two components ($\Delta$AIC $\approx -241.6$, $\Delta$BIC $\approx -224.0$), showing that a single Gaussian is an inadequate description of the log-radius distribution. The improvement from two to three components is far smaller in AIC ($\Delta \approx -19.6$) and essentially negligible in BIC ($\Delta \approx -2.05$), which penalizes additional parameters more heavily — indicating that a third component fits noise rather than genuine structure. The two-component model is therefore adopted as the best-supported description of the data.

It is worth being precise about what this does and does not establish. Model comparison selects between candidate densities; it does not by itself demonstrate that the selected density is bimodal. Evaluated on a fine grid, the fitted two-component mixture in fact has no interior local minimum at all, only a single maximum at $2.45\ R_\oplus$: its components are separated by less than the width required to produce a visible valley, so the mixture is unimodal. The evidence from model comparison is therefore that one Gaussian will not do, which is consistent with two underlying populations but is not itself evidence of a gap. Evidence for an actual gap must come from the non-parametric estimate in Section 3.2.

The fitted two-component mixture identifies a smaller population centered at $R = 1.36\ R_\oplus$ (weight $43.7\%$) and a larger population centered at $R = 2.51\ R_\oplus$ (weight $56.3\%$), consistent with the rocky super-Earth and sub-Neptune populations reported by Fulton et al. (2017).

## 3.2 Gap Location: Kernel Density Estimation and Bootstrap

Since the fitted mixture has no interior minimum of its own, locating a gap requires an estimator that is not constrained to a two-Gaussian shape. A non-parametric kernel density estimate (KDE) of the raw data

serves this purpose. This two-stage approach — GMM to establish that a single Gaussian will not describe the distribution, KDE to locate a valley if one is present — follows the structure of the methodology of Fulton et al. (2017).

A KDE fitted with the data-driven default bandwidth (Scott's rule, factor $= 0.208$) identifies a local minimum at $R = 1.76\ R_\oplus$, flanked by local maxima at $R = 1.58\ R_\oplus$ and $R = 2.46\ R_\oplus$. The location is stable under variation of the smoothing bandwidth: the minimum falls at $1.76\ R_\oplus$ under Scott's rule, $1.78\ R_\oplus$ at factor $0.18$, and $1.80\ R_\oplus$ at factor $0.15$, all well within the bootstrap interval reported below. The dip is therefore a feature of the data rather than an artifact of the bandwidth choice.

Its amplitude, however, is modest. The estimated density at the minimum is $1.599$ against $1.638$ at the adjacent lower-radius peak, a suppression of about $2\%$ — far shallower than the deficit reported by Fulton et al. (2017) for the California-Kepler Survey sample. The feature is stable in location but weak in depth, a distinction that matters for the multimodality tests of Section 3.3.

It is worth noting explicitly that the KDE and the fitted mixture disagree about modality on the same data: the KDE has two local maxima, the mixture one. This is not a contradiction but a difference in smoothing. A two-component Gaussian mixture can only represent structure expressible as two Gaussians, and fitting it by maximum likelihood favours a configuration that reproduces the overall skew; the KDE at Scott's bandwidth smooths less and follows local features the mixture cannot express. Whether such a feature is signal or sampling noise is precisely the question the formal tests of Section 3.3 address.

To quantify uncertainty, the gap location was re-estimated across 500 bootstrap resamples of the main sample. A local minimum was successfully identified in 436 of 500 resamples (87.2%); the remaining resamples produced no interior minimum within the search window, reflecting natural sampling variability rather than a methodological failure. The resulting estimate is:

$$R_{\text{gap}} = 1.76\ R_\oplus, \quad 95\%\ \text{CI} = [1.67, 1.83]\ R_\oplus.$$

As a sensitivity check, this bootstrap procedure was repeated on four independently filtered subsamples, summarized in Table 3.

Table 3: Sensitivity of the gap location estimate to filtering choices. Each row uses 300 bootstrap resamples; the headline estimate in the text uses 500.

| Subset | $N$ | Mean gap location ($R_\oplus$) | 95% CI |
|---|---|---|---|
| Main sample | 2587 | 1.761 | [1.678, 1.833] |
| Transit only | 2562 | 1.750 | [1.662, 1.833] |
| Period $< 100$ days | 2502 | 1.757 | [1.661, 1.833] |
| Relative error $< 10\%$ | 1612 | 1.737 | [1.620, 1.811] |

All four estimates fall within a $0.03\ R_\oplus$ range of one another, with substantially overlapping confidence intervals. This indicates that the gap location is robust to reasonable variation in the filtering protocol — restricting to transit-detected planets, shorter orbital periods, or stricter measurement precision does not meaningfully shift the estimate away from $\sim 1.75$–$1.76\ R_\oplus$.

### 3.3 Multimodality Testing

As a further, independent check, two formal statistical tests of unimodality were applied: Hartigan's dip test and Silverman's critical bandwidth test. Both test the null hypothesis that the underlying distribution is unimodal.

- **Hartigan's dip test**: dip statistic $= 0.0073$, $p = 0.524$.

- **Silverman's critical bandwidth test** (smoothed bootstrap, 1,000 iterations): critical bandwidth $h_{\text{crit}} = 0.0445$, $p = 0.028$.

The two tests disagree. Silverman's test rejects unimodality at the conventional $\alpha = 0.05$ level; Hartigan's does not. Because the tests are applied to identical data, the disagreement must lie in what their statistics measure.

The dip statistic is the largest vertical distance between the empirical distribution function and the closest unimodal distribution function. It is therefore governed by how *deep* a valley is, and at a suppression of $2\%$ the present valley barely registers. Silverman's statistic is the amount of smoothing required before a second mode disappears, which responds to whether a distinct second mode exists at all, largely independently of the depth of the intervening valley. A shallow but genuine second mode is exactly the configuration on which the two tests part company.

That the dip test's null result reflects insensitivity rather than absence can be checked directly rather than assumed. Simulating populations of $n = 2{,}587$ from two-component mixtures in $\log_{10}$(radius) and applying the dip test: for components at $1.3$ and $2.4$ $R_\oplus$ with standard deviation $0.08$ dex — narrow enough to produce a pronounced valley, and chosen to illustrate a clearly bimodal reference rather than fitted to any published sample — unimodality is rejected in $100\%$ of realizations. Widening the components to $0.10$ dex drops this to about $15\%$, to $0.12$ dex to none, and the configuration actually fitted to this sample is likewise never detected. The dip test at this sample size would therefore recover a pronounced gap with near-certainty and has essentially no power against a valley as shallow as the one observed.

Taken together, the two results are consistent with a second mode that is real but shallow. The most likely reason it is shallower here than in the California-Kepler Survey is sample construction: the composite archive sample pools heterogeneous literature sources and admits no survey-completeness correction, whereas the California-Kepler Survey measurement was made on a single homogeneous sample with uniformly precise radii.

A note on implementation. Silverman's test requires the bootstrap to draw from the smoothed null density at $h_{\text{crit}}$, with the resampled values rescaled to preserve the variance of the data. Resampling the observed values directly is not equivalent: duplicated points and the gaps between them generate spurious modes at a small bandwidth, which inflates the proportion of multimodal bootstrap samples and makes the test severely conservative. An earlier version of this analysis used the latter procedure and obtained $p = 0.54$; the value reported above uses the smoothed bootstrap, which returns $p \approx 0.5$ on simulated unimodal data of the same size and variance and $p \approx 0$ on a simulated clearly bimodal sample.

### 3.4 Summary

Taken together, these three analyses support a nuanced but coherent conclusion. Model comparison establishes that a single Gaussian cannot describe the distribution, though the preferred two-component mixture is itself unimodal. The non-parametric estimate locates a density minimum at $1.75$–$1.76$ $R_\oplus$ whose *position* is robust — stable across smoothing bandwidths and across four independent filtering choices — but whose *depth* is small, roughly $2\%$ below the adjacent peak. Of the two formal tests, Silverman's rejects unimodality ($p = 0.028$) and Hartigan's does not ($p = 0.52$), a split the power simulation attributes to the shallowness of the valley and to the different quantities the two statistics measure.

The defensible summary is a bandwidth-stable, filter-robust second mode near $1.76$ $R_\oplus$, separated by a valley that is real but shallow — considerably weaker in amplitude than the feature reported by Fulton et al. (2017). The location is well determined and the existence of the second mode is supported by one of the two formal tests; a gap of comparable strength to the literature one is not present in this sample. This combination of evidence is reported transparently rather than selectively, consistent with the evidentiary standard applied throughout this study.

# 4 Classification: Predicting Gap Membership

## 4.1 Class Definition and Predictor Selection

To identify which planetary and stellar properties are associated with lying above or below the radius gap, planets in the main sample were labeled using the bootstrap point estimate of Section 3.2 ($R_{\text{gap}} = 1.76\ R_\oplus$): planets with $R > 1.76\ R_\oplus$ were labeled *above-gap* (1,570 planets, $60.7\%$) and the remainder *under-gap* (1,017 planets, $39.3\%$). This threshold-based labeling introduces a moderate class imbalance, addressed during model training via class weighting rather than resampling.

Nine candidate predictors were initially considered: orbital period, orbital eccentricity, insolation, stellar effective temperature, stellar radius, stellar mass, stellar metallicity, system distance, and number of planets in the system. Radius-derived quantities (e.g., `pl_radj`, planet density) were excluded by design, since they would trivially reconstruct the class label rather than provide independent predictive information. Orbital eccentricity was further excluded due to substantial missingness (516 of 2,587 planets, $20\%$), which would have disproportionately reduced the usable sample; the remaining eight predictors had negligible to modest missingness. Restricting to complete cases across these eight predictors yielded 2,289 planets ($60.2\%$ above-gap), which were split into stratified training ($n = 1716$) and test ($n = 573$) sets, each preserving the original class proportions.

Two further preprocessing steps were applied before model training. First, the strongly right-skewed predictors (orbital period, insolation, distance) were $\log_{10}$-transformed. Second, all predictors were standardized (zero mean, unit variance) using a scaler fit exclusively on the training set and applied unchanged to the test set, preventing information leakage. Standardization places logistic regression coefficients on a common scale, which is necessary — though, as Section 4.3 discusses, not sufficient — for comparing their magnitudes across predictors; the random forest classifier, whose splits are scale-invariant, was trained on the unscaled predictors instead.

## 4.2 Model Training and Evaluation

Two classifiers were trained: logistic regression, providing an interpretable linear baseline, and a random forest (300 trees), capable of capturing non-linear relationships and interactions. Both used balanced class weighting to counteract the $60/40$ imbalance. Table 4 summarizes test-set performance.

Table 4: Classification performance on the held-out test set ($n = 573$).

| Model | ROC-AUC | Precision (under) | Recall (under) | Recall (above) |
|---|---|---|---|---|
| Logistic Regression | 0.770 | 0.62 | 0.73 | 0.71 |
| Random Forest | 0.790 | 0.70 | 0.57 | 0.84 |

Both models achieve moderate, clearly above-chance predictive power (ROC-AUC $\approx 0.77$–$0.79$, against a random-guess baseline of $0.5$), confirming that orbital and stellar properties alone — without any direct information about radius — carry real information about gap membership. The two models differ in their error trade-off: logistic regression balances recall fairly evenly across classes, while the random forest achieves a higher overall ROC-AUC but at the cost of weaker recall on the under-gap class, remaining somewhat biased toward the majority class despite weighting. Neither model is unambiguously superior; both are reported as complementary evidence of a similar underlying signal.

To confirm these results were not an artifact of a single train-test split, five-fold stratified cross-validation was performed on the training set. Mean ROC-AUC was $0.761$ (SD $0.018$) for logistic regression and $0.811$ (SD $0.009$) for random forest, consistent with the test-set figures and confirming that the observed predictive power is stable rather than a product of a favorable split.

### 4.3 Feature Importance

Table 5 reports standardized logistic regression coefficients and random forest feature importances, both sorted by magnitude.

Table 5: Predictor importance: logistic regression coefficients (standardized) and random forest feature importance.

| Predictor (LR) | Coefficient | Predictor (RF) | Importance |
|---|---|---|---|
| pl_orbper | +0.692 | pl_orbper | 0.224 |
| pl_insol | −0.533 | pl_insol | 0.179 |
| st_mass | +0.492 | sy_dist | 0.145 |
| st_met | +0.237 | st_teff | 0.109 |
| sy_pnum | −0.201 | st_rad | 0.107 |
| sy_dist | −0.169 | st_met | 0.101 |
| st_teff | −0.059 | st_mass | 0.098 |
| st_rad | +0.021 | sy_pnum | 0.036 |

Both models agree that orbital period and insolation are the dominant predictors, together capturing the great majority of predictive signal. Planets on longer orbits with lower insolation are more likely to lie above the gap, directly consistent with mass-loss theory: close-in, highly irradiated planets are expected to lose more of their primordial atmosphere and end up smaller. Stellar mass and metallicity contribute secondary positive associations with above-gap status, plausibly reflecting more gas-rich protoplanetary disks around more massive, metal-rich stars. The number of planets in the system shows a negative association with above-gap status, consistent with the "peas in a pod" tendency for planets in the same system to share similar (and often smaller) sizes. System distance emerges as moderately important in both models but is interpreted as a detection-bias artifact — more distant systems have less precisely characterized parameters — rather than a physical driver of planet size.

These coefficients should be read as a group rather than as a ranking. Orbital period and insolation are physically near-redundant, since insolation scales as $L_\star/a^2$ while $a \propto P^{2/3}$; measured on the model sample they correlate at $r = -0.74$, and their variance inflation factors (21.5 and 26.9, with a further 22.7 for stellar mass and 17.6 for effective temperature) lie well above the conventional threshold of 10. Individual coefficients within this set are therefore not separately identifiable. The instability is not merely theoretical: permutation importance computed on the held-out set reverses the order of the top two predictors, placing insolation (0.094) above orbital period (0.054), and reduces stellar effective temperature from 0.109 to 0.002. The defensible claim is that the period–insolation axis jointly dominates the predictive signal, not that period specifically outranks insolation. Impurity-based random forest importances are additionally biased toward continuous, high-cardinality predictors, which understates the discrete sy_pnum.

### 4.4 Robustness to Threshold Choice

Since the class label depends on a single point-estimate threshold (1.76 $R_\oplus$), a robustness check re-trained both models after excluding planets falling within the union of the bootstrap 95% intervals across all four filtering variants of Table 3 (1.62–1.83 $R_\oplus$, taking the lowest lower bound and the highest upper bound) — a deliberately wider exclusion than the main sample's own interval of $[1.67, 1.83]$ $R_\oplus$ — retaining only the 2,089 "clear" cases furthest from the ambiguous boundary. Results were essentially unchanged (logistic regression: $0.770 \rightarrow 0.772$; random forest: $0.790 \rightarrow 0.810$), with the random forest result improving slightly — consistent with boundary-adjacent planets being intrinsically harder to classify. This confirms

the classification findings are not an artifact of the specific threshold used to define the classes.

# 5 Classical Hypothesis Testing

This section reports classical, non-parametric and categorical hypothesis tests that formally test the group differences identified descriptively and via classification modeling in the preceding sections.

## 5.1 Period and Radius

Radius distributions were compared between short-period ($\leq 10$ days) and long-period ($> 10$ days) planets using the Kolmogorov-Smirnov (KS) test, which compares overall distribution shape and location, and the Mann-Whitney U test, which specifically tests for a location/rank shift. Both reject the null hypothesis of equal distributions with overwhelming significance:

- KS test: $D = 0.339$, $p = 7.77 \times 10^{-66}$
- Mann-Whitney U test: $U = 509{,}419.5$, $p = 3.89 \times 10^{-64}$

The effect is also practically large, not merely statistically detectable: median radius is $1.67\ R_\oplus$ for short-period planets versus $2.34\ R_\oplus$ for long-period planets. As a robustness extension, the sample was re-split into three period groups (short $\leq$ 5 days, medium 5–20 days, long $>$ 20 days) and compared via the Kruskal-Wallis test, which generalizes Mann-Whitney to more than two groups: $H = 381.79$, $p = 1.24 \times 10^{-83}$. This is not an independent finding but a confirmation of the same underlying relationship under a finer grouping, with the smaller $p$-value reflecting a monotonic trend (radius increasing with period) captured more sharply by the additional group.

## 5.2 Categorical Associations with Gap Position

Two categorical association tests examined whether gap position (above/under) relates to system-level properties. First, a chi-square test of independence tested association with multi-planet system membership:

$$\chi^2 = 13.18, \quad p = 2.83 \times 10^{-4}, \quad \text{dof} = 1.$$

All expected cell counts exceeded 5, satisfying the test's validity requirement. The null hypothesis of independence is rejected: planets in multi-planet systems are modestly over-represented below the gap (535 observed versus 489.4 expected) and under-represented above it (710 versus 755.6 expected), consistent with the "peas in a pod" tendency for planets within a system to share similar, often smaller, sizes.

Second, association with discovery method was tested. Because discovery-method counts are heavily imbalanced after filtering (2,562 Transit-detected planets versus 25 combined Radial Velocity and Transit Timing Variation detections), a chi-square test would be unreliable due to small expected cell counts; Fisher's exact test was used instead, comparing Transit against all other methods combined: odds ratio $= 1.22$, $p = 0.68$. This result is not significant, and the null hypothesis of independence is not rejected. Given the very small non-Transit sample, this should be read as an inconclusive, underpowered result rather than positive evidence of no association.

## 5.3 Group Differences in the Candidate Predictors

Mann-Whitney U tests compared above-gap and under-gap planets on four continuous variables: insolation, stellar mass, stellar metallicity, and orbital eccentricity. The first three were among the predictors used by

the classification models of Section 4; eccentricity was excluded there for missingness, and is included here as a candidate predictor those models could not assess. All four differences were statistically significant (Table 6), and all four remain so after Holm correction for the multiple tests reported in this section.

These tests use the same sample, the same variables and the same class labels as the classification models, so they restate the model-based associations in classical form rather than corroborating them from independent evidence.

Table 6: Mann-Whitney U tests comparing above-gap and under-gap planets.

| Predictor | $p$-value | Median (under-gap) | Median (above-gap) |
|---|---|---|---|
| pl_insol | $3.21 \times 10^{-47}$ | 166.0 | 47.3 |
| st_mass | $1.50 \times 10^{-9}$ | 0.854 | 0.900 |
| st_met | $1.10 \times 10^{-3}$ | $-0.01$ | 0.00 |
| pl_orbeccen | $3.79 \times 10^{-7}$ | 0.0 | 0.0 |

Insolation shows by far the strongest effect, with under-gap planets receiving substantially more stellar irradiation, consistent with closer-in planets losing more atmosphere. Stellar mass and metallicity show smaller but significant positive associations with above-gap status. Orbital eccentricity is a notable methodological case: despite a highly significant test result, both groups share an identical median of exactly zero. This is not a contradiction — many planets default to an assumed circular orbit when eccentricity is not well constrained, so the significant Mann-Whitney result reflects a difference in the upper tail of the rank distribution rather than a shift in the typical (median) value, illustrating that a significant rank-based test does not always imply a shift in central tendency.

### 5.4 Summary

Every hypothesis test in this section is consistent with the descriptive and model-based findings of the preceding sections. The period-radius relationship is confirmed with overwhelming significance by three complementary tests (treated as one robustly confirmed finding rather than three independent results). Gap position is significantly associated with multi-planet system membership but shows no significant association with discovery method, the latter likely reflecting limited statistical power rather than a genuine absence of effect. All four continuous predictors tested show statistically significant differences between above-gap and under-gap groups, surviving Holm correction. Because these tests reuse the sample, variables and labels of the classification analysis, they restate the model-based feature importance results in classical form rather than confirming them independently.

## 6 Bayesian Analysis: Beta-Binomial Model

As a complementary approach to the classical tests in Section 5, this section presents a Bayesian analysis of the same underlying question: does the proportion of above-gap ("sub-Neptune-like") planets differ between short-period and long-period planets? Where classical hypothesis testing yields a binary reject/fail-to-reject decision and a $p$-value, the Bayesian approach yields a full probability distribution over the parameter of interest, allowing direct probabilistic statements such as "there is a 95% probability the true proportion lies in this range."

### 6.1 Model

Planets were again split into short-period ($\leq 10$ days, $n = 1{,}415$, of which 644 above-gap) and long-period ($> 10$ days, $n = 1{,}172$, of which 926 above-gap) groups. For each group, the number of above-gap planets was modeled as a Binomial outcome with unknown success probability $p$, and a Beta prior was placed on $p$. The Beta distribution is the conjugate prior for the Binomial likelihood, meaning the posterior is again a Beta distribution with parameters updated by simple addition:

$$p \sim \text{Beta}(a, b) \quad \Longrightarrow \quad p \mid \text{data} \sim \text{Beta}(a + s,\ b + f),$$

where $s$ and $f$ denote the observed number of successes (above-gap planets) and failures (under-gap planets) in the group. Starting from an uninformative uniform prior, $\text{Beta}(1, 1)$, the posteriors are:

$$p_{\text{short}} \mid \text{data} \sim \text{Beta}(645, 772), \qquad p_{\text{long}} \mid \text{data} \sim \text{Beta}(927, 247).$$

### 6.2 Posterior Estimates

Table 7 summarizes the resulting posterior distributions.

Table 7: Posterior estimates of the proportion above the gap, by period group (Beta(1,1) prior).

| Group | Posterior mean | Posterior median | 95% credible interval |
|---|---|---|---|
| Short-period | 0.455 | 0.455 | [0.429, 0.481] |
| Long-period | 0.790 | 0.790 | [0.766, 0.812] |

The two 95% credible intervals do not overlap at all, providing strong evidence that the true proportions genuinely differ between groups. Unlike a frequentist confidence interval, a Bayesian credible interval admits a direct probabilistic interpretation: given the observed data, there is a 95% probability that the true proportion lies within the stated range.

### 6.3 Posterior Difference

To directly quantify the difference between groups, $\Delta = p_{\text{long}} - p_{\text{short}}$ was estimated via Monte Carlo simulation: 100,000 samples were drawn independently from each posterior, and their difference computed for each pair. This yields:

$$\text{mean } \Delta = 0.335, \qquad 95\%\ \text{CI} = [0.299, 0.369], \qquad P(p_{\text{long}} > p_{\text{short}} \mid \text{data}) > 0.9999.$$

The credible interval for $\Delta$ excludes zero entirely, and essentially all posterior mass supports $p_{\text{long}} > p_{\text{short}}$. The probability is reported as $> 0.9999$ rather than as exactly unity because it is a Monte Carlo estimate from $100{,}000$ draws, none of which fell below zero.

This establishes that long-period planets in the present sample are more often above the gap. Attributing that difference to atmospheric mass loss requires separately excluding period-dependent detection bias, which Section 7.3 addresses.

### 6.4 Robustness Checks

Four checks were performed to confirm this conclusion does not depend on arbitrary modeling choices. First, **prior sensitivity** was tested using three priors of increasing strength: the original uniform $\text{Beta}(1, 1)$, a mildly informative $\text{Beta}(2, 2)$, and a strongly "skeptical" $\text{Beta}(20, 20)$ that pulls hard toward $p = 0.5$.

Results were nearly identical across all three (mean $\Delta = 0.335$, $0.334$, and $0.324$ respectively; $P(\Delta > 0) > 0.9999$ in every case), confirming the conclusion is driven by the data rather than the choice of prior.

Second, a **posterior predictive check** simulated 100,000 replicated datasets — drawing $p$ from each posterior and generating a hypothetical count of above-gap planets via a Binomial draw — and compared these to the observed counts (644 and 926). In both groups the observed count fell squarely within the bulk of the simulated distribution. This confirms that the conjugate update was implemented correctly, but it is weak evidence of model adequacy: because each posterior was conditioned on precisely the count being predicted, replicated counts will centre on the observed values whether or not the Binomial likelihood is appropriate, so the check cannot detect misfit in the likelihood itself.

Third, **sensitivity to the period threshold** used to define "short" versus "long" was tested by repeating the full analysis with thresholds of 5, 10, and 20 days (Table 8).

Table 8: Sensitivity of the posterior difference to the period threshold definition.

| Threshold | $p_{\text{short}}$ | $p_{\text{long}}$ | Mean $\Delta$ | 95% CI | $P(\Delta > 0)$ |
|---|---|---|---|---|---|
| 5 days | 0.320 | 0.735 | 0.414 | [0.375, 0.452] | $> 0.9999$ |
| 10 days | 0.455 | 0.790 | 0.334 | [0.299, 0.369] | $> 0.9999$ |
| 20 days | 0.544 | 0.822 | 0.276 | [0.238, 0.313] | $> 0.9999$ |

Across all three thresholds, $P(\Delta > 0) > 0.9999$ and the credible interval never approaches zero. The magnitude of the effect shrinks somewhat as the threshold widens (since the "short" group then includes planets progressively farther from the star), but the direction and certainty of the effect are completely stable, confirming the finding is not an artifact of one arbitrary threshold choice.

Fourth, the **independence assumption** underlying the Binomial likelihood was tested directly. Planets in the same system do not have independent radii: 1,089 of the 2,587 planets share a host with another sample member, and the "peas in a pod" effect documented in Section 2.2 implies positive within-system correlation, so the nominal credible interval could in principle be too narrow. Re-estimating $\Delta$ by bootstrapping whole host systems rather than individual planets gives $[0.299, 0.370]$, against $[0.299, 0.369]$ from the conjugate model — an interval inflation of $1.02$. Most host systems contribute a single planet to the filtered sample, and multi-planet systems typically span both the period threshold and the gap, so the clustering has little effect on a difference of proportions. The reported precision is therefore not materially overstated.

## 6.5 Summary

The Bayesian analysis provides strong, robust, and intuitively interpretable evidence — fully consistent with the classical tests of Section 5 — that long-period planets in this sample are substantially more likely to lie above the radius gap than short-period planets. This conclusion is stable across three priors of varying strength, holds under three different period-threshold definitions, and survives a cluster bootstrap that relaxes the assumption of independence between planets in the same system.

The physical interpretation is the weaker link in the chain. Photoevaporation and core-powered mass-loss both predict this pattern, but so does period-dependent transit detection bias, and the present sample does not allow the two to be separated (Section 7.3).

# 7 Discussion

## 7.1 Synthesis of Findings

Four largely independent analytical approaches converge on a consistent picture. Distributional modeling (Section 3) establishes, via AIC/BIC model comparison, that a single Gaussian cannot describe the log-radius distribution of the filtered sample, and locates a density minimum at $R \approx 1.75$–$1.76\ R_\oplus$ with a bootstrap-estimated uncertainty of roughly $\pm 0.08\ R_\oplus$, a location stable across smoothing bandwidths and across four independently constructed subsamples. The minimum is shallow, and the two formal multi-modality tests split: Silverman's critical bandwidth test rejects unimodality ($p = 0.028$) while Hartigan's dip test does not ($p = 0.52$). A power simulation at this sample size attributes the split to the modest depth of the valley, to which the dip statistic is sensitive and Silverman's is not. Classification modeling (Section 4) shows that orbital and stellar properties, deliberately excluding radius, predict gap membership with moderate but genuine skill (ROC-AUC 0.77–0.79), and identifies orbital period and insolation as the dominant predictors in both a linear and a non-linear model. Classical hypothesis testing (Section 5) confirms, with overwhelming statistical significance, that radius depends on orbital period, and that the specific predictors flagged by the classification models differ significantly between above-gap and under-gap planets. Finally, Bayesian estimation (Section 6) quantifies the period effect directly: the posterior probability that long-period planets are more likely to lie above the gap than short-period planets exceeds $0.9999$, a conclusion robust to the choice of prior, to the exact period threshold used, and to relaxing the assumption that planets in the same system are independent.

Together, these results are consistent with a physical picture in which planets closer to their host star lose more of their primordial atmosphere, shrinking below the gap, while planets farther out retain thicker envelopes and remain above it — the shared prediction of both photoevaporation and core-powered mass-loss theories. They are equally consistent, however, with period-dependent detection bias, which Section 7.3 treats as the principal limitation of this study. This study's design does not distinguish between the two mechanisms, since both predict the same qualitative period-dependence; doing so would require additional information this dataset does not directly provide, such as host-star X-ray/UV luminosity histories (relevant to photoevaporation specifically) or precise ages (relevant to the longer core-powered mass-loss timescale).

## 7.2 Comparison to the Literature

The estimated gap location (1.75–1.76 $R_\oplus$) sits within the range reported by Fulton et al. (2017) (1.5–2.0 $R_\oplus$), and the identification of orbital period and insolation as the dominant predictors of gap position is consistent with subsequent work attributing the gap to atmospheric mass-loss rather than, for example, a primordial bimodality in planet formation. The secondary association found between multi-planet system membership and gap position is consistent with the previously documented "peas in a pod" effect (Weiss et al., 2018), which this project also independently confirmed via a permutation test during initial exploration (Section 2.2).

## 7.3 Limitations

Several limitations bear on the interpretation of these results.

First, and most consequentially for the central result, the period-dependence reported in Sections 5 and 6 cannot be cleanly separated from transit detection bias. Transit signal-to-noise scales with planet radius and with the number of transits observed within a fixed baseline, so small planets on long orbits are systematically harder to detect. The effect is plainly present in the analysis sample: the fifth-percentile radius rises monotonically from $0.79\ R_\oplus$ below 3 days to $0.89$, $1.13$, $1.31$ and $1.48\ R_\oplus$ in the 3–10, 10–30, 30–100 and $> 100$ day bins respectively, while the fraction of planets smaller than $1.5\ R_\oplus$ falls from $55\%$ to

$6\%$ across the same range. A survey that preferentially fails to detect small long-period planets will produce precisely the signature reported here — a higher above-gap fraction at long period — in the complete absence of atmospheric mass loss. Distinguishing the two requires a survey-completeness correction applied within a homogeneous sample, as in Fulton et al. (2017); the composite archive table used here does not support one. The period results should accordingly be read as consistent with mass-loss theory rather than as evidence favoring it over the selection-effect explanation.

Second, the sample is not representative of the true underlying exoplanet population: the filtering protocol, combined with underlying detection biases already present in the archive, concentrates the sample heavily toward Transit-detected planets on short-to-moderate orbits around Sun-like stars. This is a necessary consequence of working with the best-characterized subset of known planets, but it means the reported relationships describe the detected and well-characterized population, not necessarily the true population of small planets in the galaxy.

Third, the PSCompPars table is a composite catalog: different columns for the same planet may originate from different literature sources and are not guaranteed to be mutually self-consistent, introducing a source of measurement heterogeneity that is difficult to quantify or fully control for.

Fourth, missingness in several predictors (most notably orbital eccentricity, excluded from the classification analysis for this reason) is not random but tied to detection method and measurement difficulty, meaning the complete-case samples used in Sections 4 and 5 may not be fully representative even of the filtered main sample.

Fifth, the classification and period-based analyses rely on a single point-estimate gap location ($1.76\ R_\oplus$) to define class labels. This risk was addressed directly by re-running the classification analysis after excluding planets within the gap's bootstrap confidence interval (Section 4.4), which left conclusions unchanged, but the underlying dependence on a chosen threshold remains a modeling choice rather than a fully data-determined boundary.

Finally, this analysis is observational and correlational throughout. While the identified predictors (period, insolation) are consistent with mass-loss theory, no causal mechanism is directly tested; the physical interpretation offered here draws on established theory rather than on evidence generated within this study alone.

# 8 Conclusion

This project investigated the exoplanet radius gap using a carefully filtered sample of 2,587 planets from the NASA Exoplanet Archive, addressing four research questions through complementary statistical approaches. The radius distribution shows a filter-robust and bandwidth-stable density minimum at $R \approx 1.76\ R_\oplus$, coinciding in location with the literature radius gap but substantially shallower than the feature reported for the California-Kepler Survey sample. Of the two formal multimodality tests, Silverman's critical bandwidth test rejects unimodality and Hartigan's dip test does not; a power simulation at this sample size shows the dip test has essentially no sensitivity to a valley this shallow, so the two results together are consistent with a second mode that is real but weak. Orbital and stellar properties, excluding radius itself, predict gap membership with moderate accuracy, with the period–insolation axis carrying most of the predictive signal, though severe collinearity within that pair prevents the two from being separately ranked. This period-dependence was quantified through classical hypothesis testing and through Bayesian estimation, both providing strong, threshold-robust, and prior-robust evidence that planets on longer orbits are more likely to retain larger, gas-enveloped radii than their close-in counterparts.

Taken together, these findings are consistent with atmospheric mass-loss — whether driven by stellar irradiation, internal planetary heat, or both — as the physical process shaping the radius gap. They do not, however, establish it over the competing explanation of period-dependent detection bias, which predicts the

same relationship and which a composite archive sample without completeness corrections cannot rule out. Doing so is the natural next step for this analysis, and would require a homogeneous, completeness-corrected sample of the kind used by Fulton et al. (2017).

## References


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